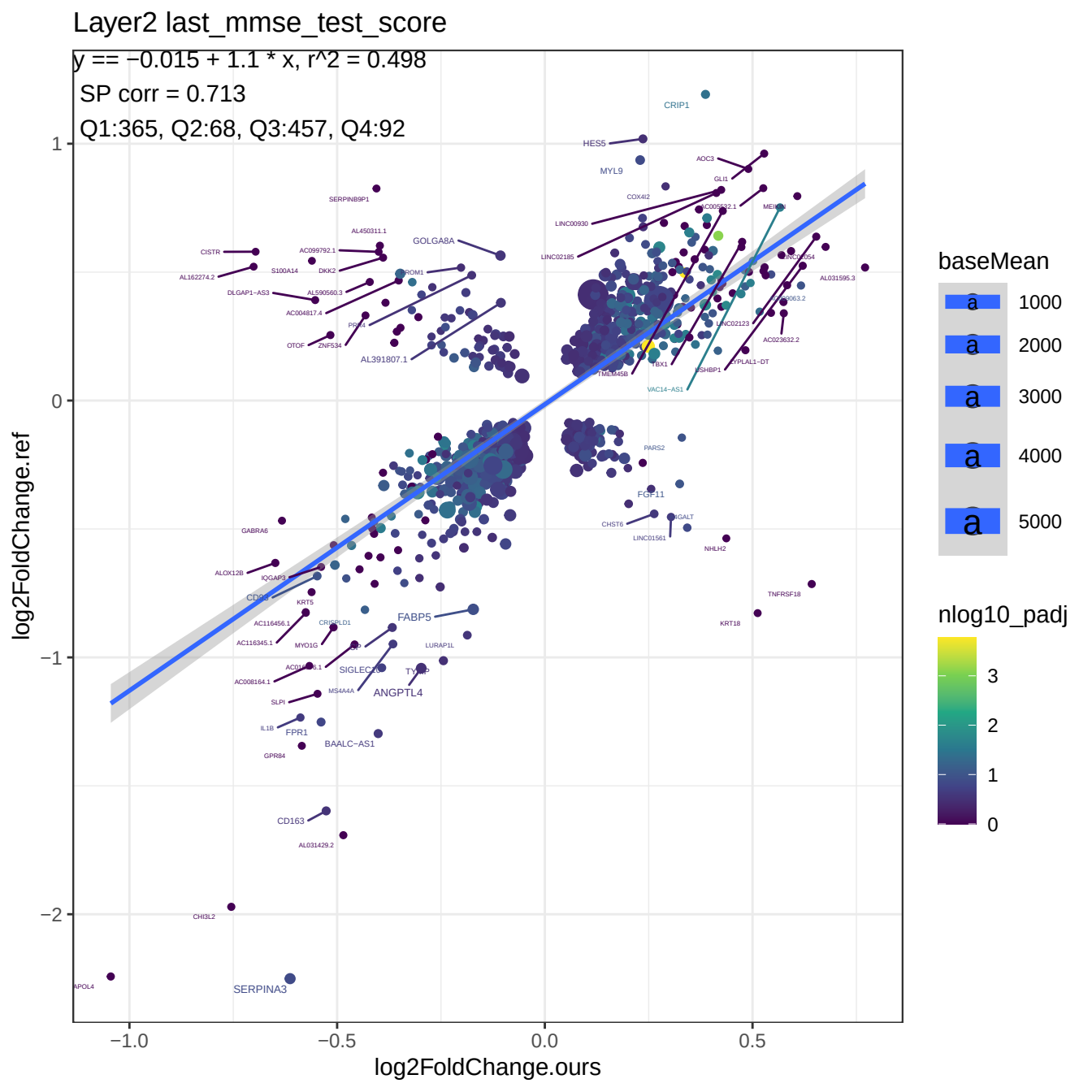
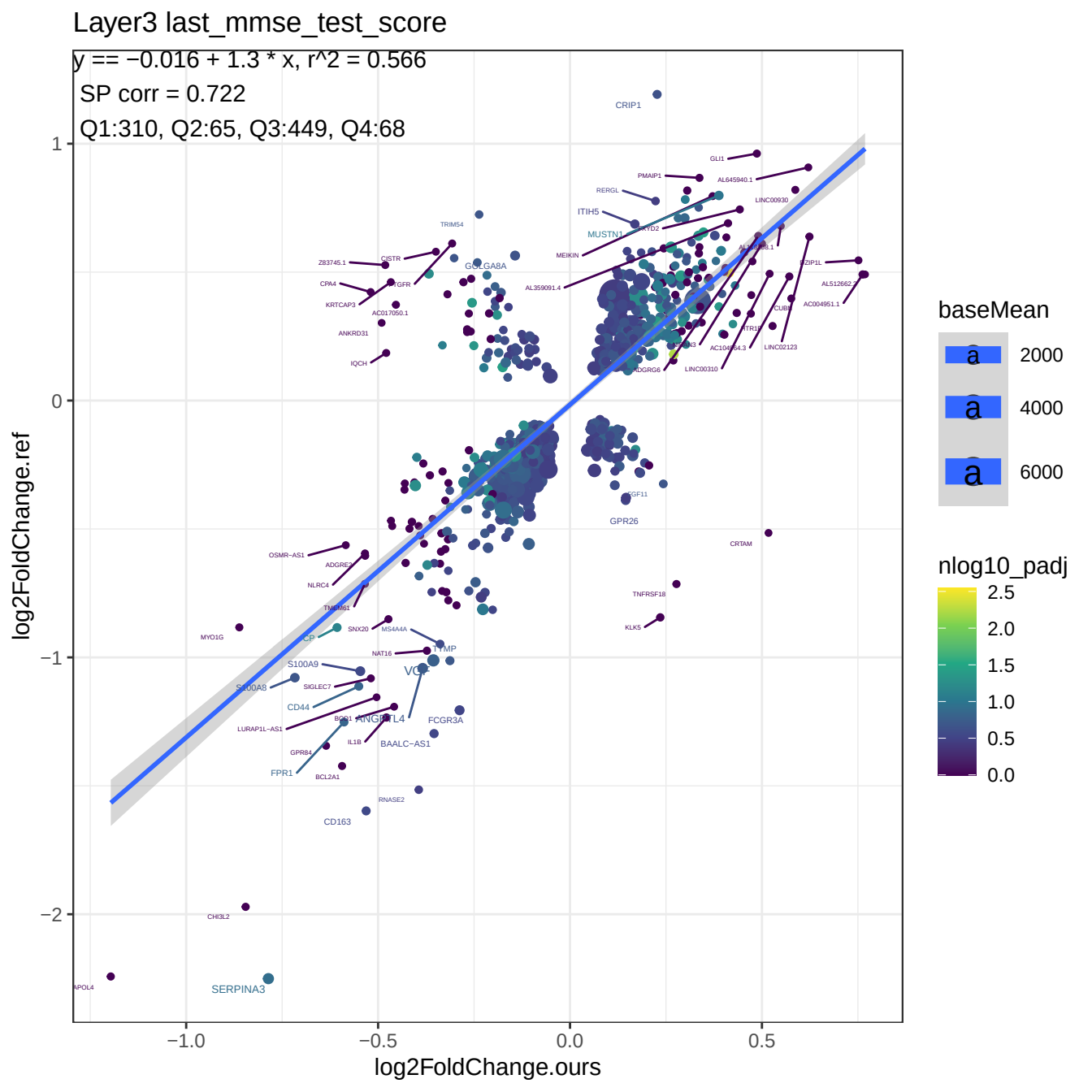
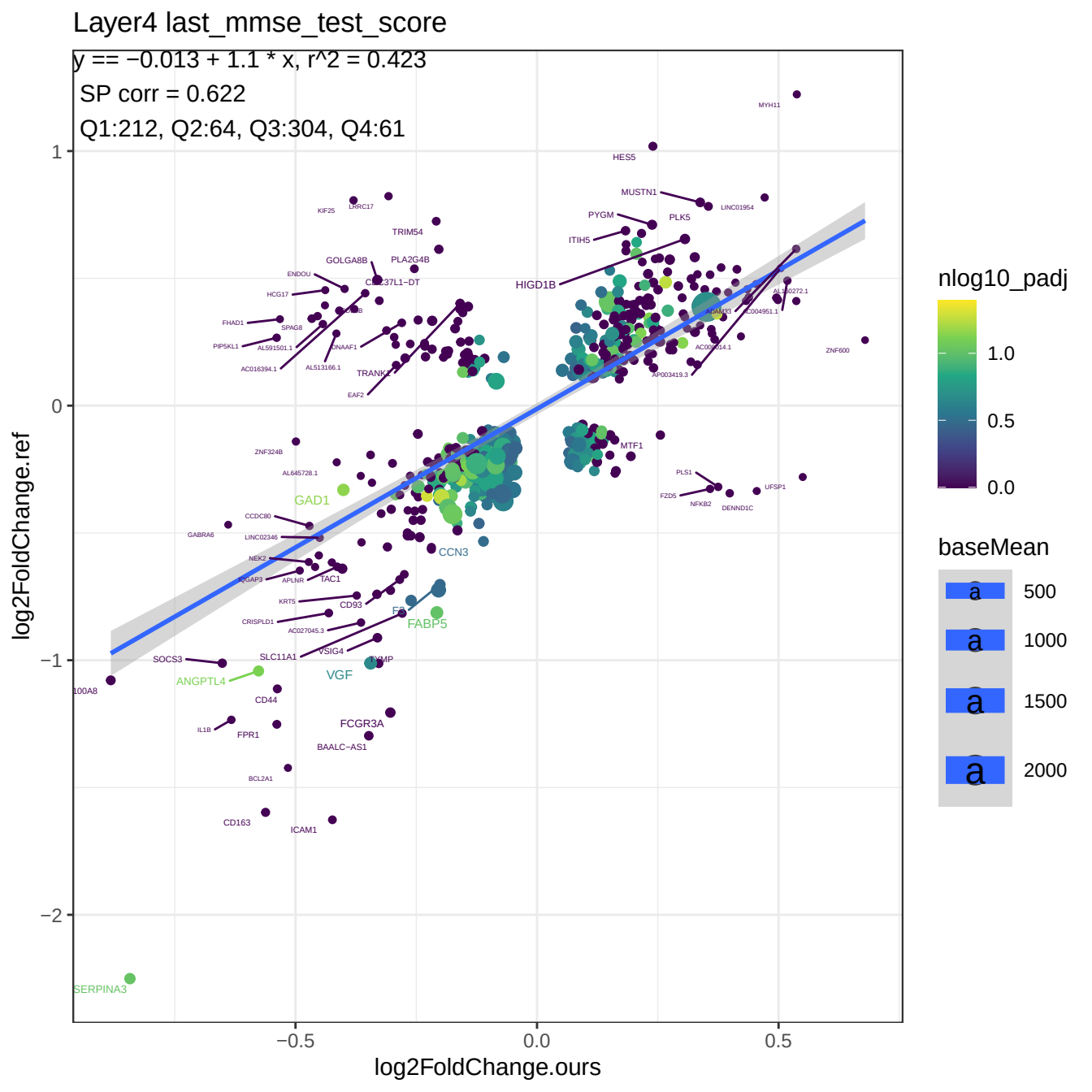
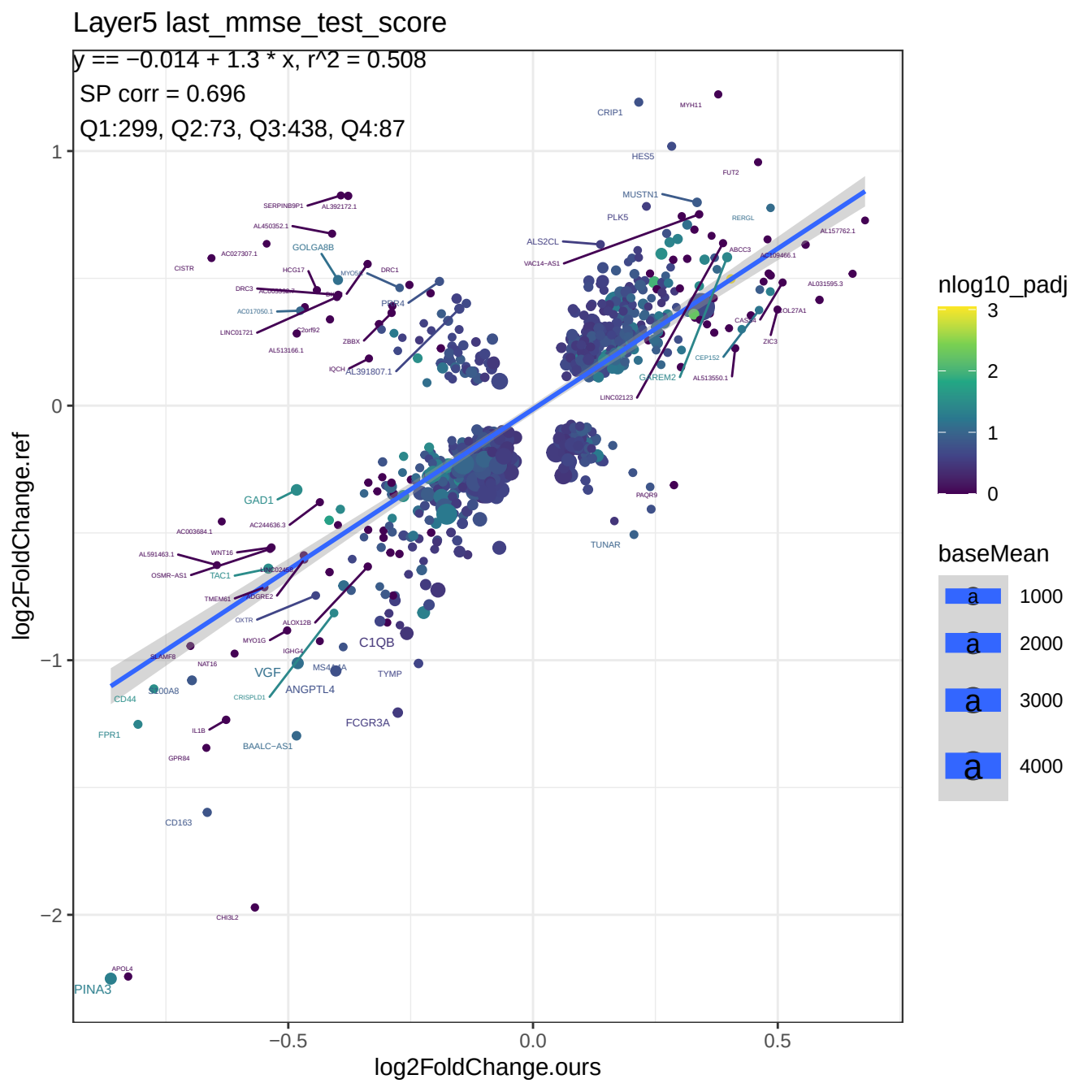
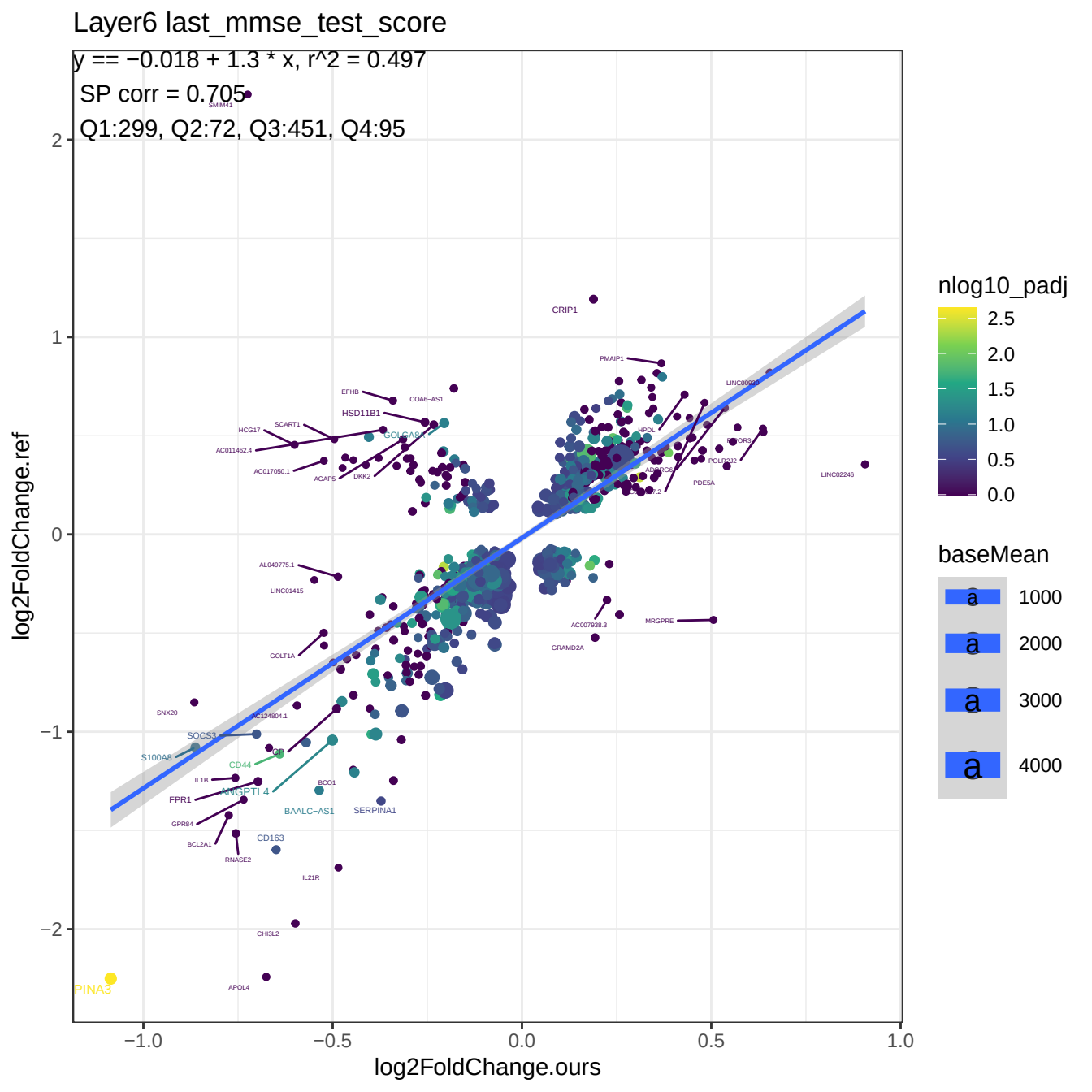
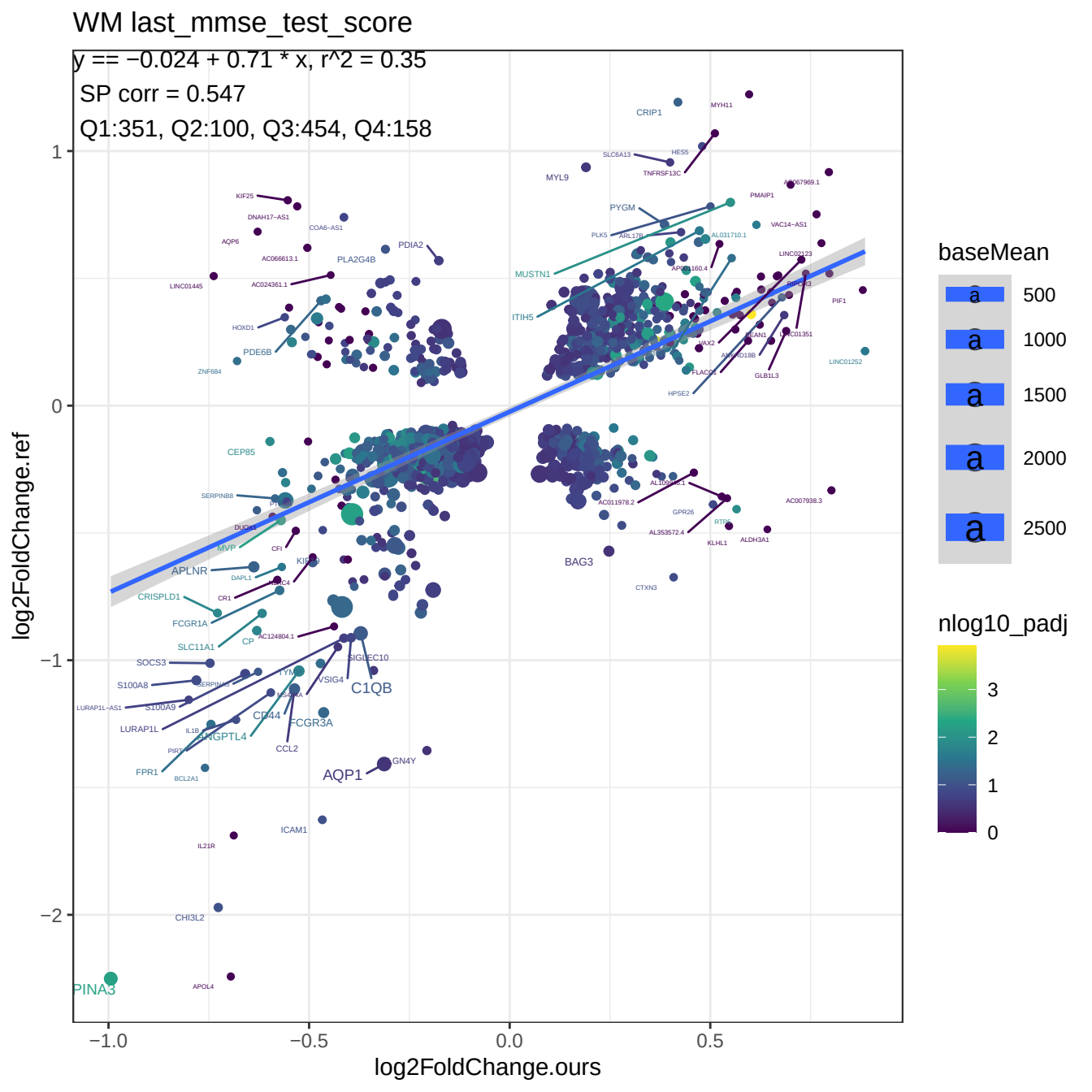

Layer2 last_mmse_test_score
y == −0.015 + 1.1 * x, r^2 = 0.498
SP corr = 0.713
Q1:365, Q2:68, Q3:457, Q4:92
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
baseMean
1000
2000
3000
4000
5000
nlog10_padj
3
2
1
0
CRIP1
HES5
MYL9
AOC3
GLI1
COX4I2
LINC00930
LINC02185
MEIS1
AL031595.3
AC023632.2
LINC02123
TYPLAL1−DT
TBX1
TMEM45B
VAC14−AS1
PARS2
FGF11
CHST6
B3GALT
LINC01561
NHLH2
TNFRSF18
KRT18
SERPINB9P1
AL450311.1
CISTR
AC099792.1
S100A14
DKK2
AL162274.2
DLGAP1−AS3
AL590560.3
AC004817.4
PRR4
OTOF
ZNF534
GOLGA8A
PROM1
AL391807.1
GABRA6
ALOX12B
IQGAP3
CD93
KRT5
AC116456.1
AC116345.1
MYO1G
CRISPLD1
FABP5
LURAP1L
SIGLEC10
TXNIP
ANGPTL4
MS4A4A
SLPI
IL1B
FPR1
GPR84
BAALC−AS1
CD163
AL031429.2
CHI3L2
APOL4
SERPINA3


Layer3 last_mmse_test_score
y == −0.016 + 1.3 * x, r^2 = 0.566
SP corr = 0.722
Q1:310, Q2:65, Q3:449, Q4:68
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
baseMean
a
2000
4000
6000
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0
CRIP1
GLI1
PMAIP1
AL645940.1
RERGL
LINC00930
ITIH5
TRIM54
MUSTN1
MEIKIN
GOLGA8A
Z83745.1
CISTR
CPA4
KRTCAP3
AC017050.1
ANKRD31
IQCH
AL359091.4
ZIP1L
AL512662.1
AC004951.1
LINC02123
LINC00310
AC104984.3
PDGRG6
FGF11
GPR26
CRTAM
TNFRSF18
KLK5
OSMR−AS1
ADGRE3
NLRC4
TMEM61
MYO1G
SNX20
MS4A4A
CP
NAT16
TYMP
VCP
S100A9
S100A8
SIGLEC7
CD44
BCS1
ANGPTL4
FCGR3A
LURAP1L−AS1
IL1B
BAALC−AS1
GPR84
FPR1
BCL2A1
RNASE2
CD163
CHI3L2
APOL4
SERPINA3


Layer4 last_mmse_test_score
y == −0.013 + 1.1 * x, r^2 = 0.423
SP corr = 0.622
Q1:212, Q2:64, Q3:304, Q4:61
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−0.5
0.0
0.5
nlog10_padj
1.0
0.5
0.0
baseMean
500
1000
1500
2000
MYH11
HES5
MUSTN1
LINC01954
PYGM
PLK5
ITIH5
HIGD1B
KIF25
LRRC17
TRIM54
GOLGA8B
PLA2G4B
ENDOU
HCG17
FHAD1
SPAG8
PIP5KL1
AL591501.1
AC016394.1
AL513166.1
EAF2
ZNF324B
AL645728.1
GAD1
CCDC80
GABRA6
LINC02346
NEK2
ARHGAP3
APLNR
TAC1
KRT5
CD93
CRISPLD1
AC027045.3
VSIG4
SOCS3
SLC11A1
ANGPTL4
VGF
CD44
IL1B
FPR1
FCGR3A
BAALC−AS1
BCL2A1
CD163
ICAM1
SERPINA3
MTF1
PLS1
FZD5
NFKB2
DENND1C
UFSP1
ZNF600
AP003419.3
CCN3
FABP5


Layer5 last_mmse_test_score
y == −0.014 + 1.3 * x, r^2 = 0.508
SP corr = 0.696
Q1:299, Q2:73, Q3:438, Q4:87
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−0.5
0.0
0.5
CRIP1
MYH11
HES5
FUT2
MUSTN1
PLK5
RERGL
AL157762.1
ALS2CL
VAC14−AS1
ABCC3
AC109466.1
AL031595.3
SERPINB9P1
AL392172.1
AL450352.1
GOLGA8B
AC027307.1
CISTR
HCG17
MYO5C
DRC1
DRC3
PDE4A
AC017050.1
LINC01721
C2orf92
ZBBX
AL513166.1
IQCH
AL391807.1
COL27A1
CASS4
ZIC3
CEP152
GAREM2
AL513550.1
LINC02123
PAQR9
TUNAR
GAD1
AC244636.3
AC003684.1
AL591463.1
WNT16
OSMR−AS1
TAC1
AC024937.2
TMEM81
ADGRE2
OXTR
ALOX12B
MYO1G
IGHG4
SLAMF8
NAT16
VGF
MS4A4A
TYMP
C1QB
ANGPTL4
CRISPLD1
CD44
S100A8
FCGR3A
FPR1
IL1B
BAALC−AS1
GPR84
CD163
CHI3L2
APOL4
PINA3
nlog10_padj
3
2
1
0
baseMean
a
1000
2000
3000
4000


Layer6 last_mmse_test_score
y == −0.018 + 1.3 * x, r^2 = 0.497
SP corr = 0.705
Q1:299, Q2:72, Q3:451, Q4:95
log2FoldChange.ref
log2FoldChange.ours
2
1
0
−1
−2
−1.0
−0.5
0.0
0.5
1.0
nlog10_padj
2.5
2.0
1.5
1.0
0.5
0.0
baseMean
1000
2000
3000
4000
SMIM41
CRIP1
PMAIP1
LINC00936
EFHB
COA6−AS1
HSD11B1
HCG17
SCART1
GOLGA8A
AC011462.4
AC017050.1
AGAP5
DKK2
HPDL
RNF223
POLR2J2
ADAMTS6
PDE5A
LINC02246
AL049775.1
LINC01415
AC007938.3
MRGPRE
GRAMD2A
GOLT1A
SNX20
AC124804.1
SOCS3
S100A8
CD44
IL1B
FPR1
ANGPTL4
BCO1
BAALC−AS1
SERPINA1
GPR84
BCL2A1
CD163
RNASE2
IL21R
CHI3L2
APOL4
PINA3


WM last_mmse_test_score
y == −0.024 + 0.71 * x, r^2 = 0.35
SP corr = 0.547
Q1:351, Q2:100, Q3:454, Q4:158
log2FoldChange.ref
log2FoldChange.ours
1
0
−1
−2
−1.0
−0.5
0.0
0.5
baseMean
500
1000
1500
2000
2500
nlog10_padj
3
2
1
0
CRIP1
MYH11
SLC6A13
HES5
TNFRSF13C
MYL9
PYGM
PMAIP1
VAC14−AS1
PLK5
ARL17B
AL031710.1
LINC02123
MUSTN1
ITIH5
PIF1
LINC01351
LINC01252
GLB1L3
HPSE2
KIF25
DNAH17−AS1
COA6−AS1
AQP6
AC066613.1
PDIA2
PLA2G4B
LINC01445
AC024361.1
HOXD1
PDE6B
ZNF684
CEP85
SERPINB8
DUOX1
MVP
CFI
APLNR
DAPL1
CRISPLD1
CR1
FCGR1A
CP
AC124804.1
SLC11A1
SOCS3
S100A8
S100A9
LURAP1L−AS1
LURAP1L
IL1B
PIRT
FPR1
BCL2A1
TYMP
SERPINB2
VSIG4
SIGLEC10
C1QB
CD44
FCGR3A
ANGPTL4
CCL2
AQP1
CGN4Y
ICAM1
IL21R
CHI3L2
APOL4
PINA3
BAG3
CTXN3
AC011978.2
GPR26
AL353572.4
KLHL1
ALDH3A1
AC007938.3